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Ultrasonic cavitation monitoring by acoustic noise power measurement

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In this paper, a new tool is proposed to carry out acoustic cavitation monitoring and to have an overview of its effects in applications. After a brief review of the cavitation characterization techniques, it is shown that cavitation noise is a suitable and accurate indicator of the cavitation activity induced in a liquid. In the first part of this study, the origin of the first spectral component of the cavitation noise is discussed. The $f/2$ and $2f$ component evolution measurement at a driving frequency around 1 MHz confirms Neppiras' ones and gives an indicator of the cavitation inception. In the second part, the cavitation noise spectrum distortion is considered as a function of the acoustic power transmitted to the liquid in order to obtain an indicator of cavitation activity. In the last part, this new tool is used to bring to the fore the hysteresis effect associated with the cavitation. An experimental correlation between cavitation noise power measurement and the sonochemical activity in an oxidization process is also presented. © 2000 Acoustical Society of America. [S0001-4966(00)01211-X]

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LIST OF SYMBOLS

R_0	initial bubble radius
R	bubble radius at any given moment
$\dot{R}$	bubble wall velocity
$\ddot{R}$	bubble wall acceleration
P_A	maximum amplitude of the acoustic pressure
P_0	equilibrium pressure in the liquid
σ	surface tension
f	driving frequency

ρ_0	density of the liquid
ω	angular frequency = $2\pi f$
n	gas polytropic index
f/q	with q integer: subharmonic of the driving frequency
pf	with p integer: harmonic of the driving frequency
pf/q	with p and q integers: ultraharmonic of the driving frequency

I. INTRODUCTION

Cavitation is defined as the formation of bubbles in a liquid subject to local pressure variations. Thus, the propagation of an acoustic wave through a liquid may induce cavitation. The effects associated with cavitation onset in a liquid cover a wide variety of physical phenomena such as luminescence, shock wave emissions, and free-radicals formation. In the past 20 years, most of the research carried out on acoustic cavitation shows that this phenomenon should be a powerful tool for many industrial applications. Cleaning systems using cavitation are now well-known. Sonochemistry, a process in which cavitation is used as a kind of chemical catalyst, has been studied for 50 years. Moreover, some authors¹⁻³ have also reported interesting results concerning the bacteriological effects of cavitation. Despite these encouraging prospects, ultrasonic cavitation still remains a laboratory tool for two main reasons.

Uncertainties concerning the mechanisms whereby cavitation enhances chemical reactivity or generates light emissions are the first reasons for limited industrial scale development of cavitation. Many works have been carried out in

order to explain the behavior of a cavitation bubble. Sehgal *et al.*,⁴ and later Suslick⁵⁻⁷ showed that collapsing bubbles may generate hot spots in the liquid. Thus, they concluded that sonochemistry and sonoluminescence arise from localized peak temperatures and pressures. Recently, Lepoint^{8,9} developed an electrical interpretation of these phenomena. He assumed that the electrical charge carried by a bubble might play a pivotal part in sonochemical mechanisms. Localized electrical fields produced by bubble distortions may generate microscopic intracavity discharges during the collapse. He concludes that the microplasma induced in the bubble is the source of sonochemical and sonoluminescent activity. The sonoluminescence emission spectrum analysis, currently receiving a great deal of attention, should rapidly provide more information about cavitation mechanisms.

The lack of cavitation bubble field probes is the second reason that limits the industrial-scale development of cavitation. An obvious reason for the complexity of cavitation characterization is the great disparity in bubble characteristics inside the cavitation field. Once it is generated, an acoustic cavitation bubble may undergo two quite different kinds of radial oscillations. It may oscillate nonlinearly during many cycles of the acoustic wave, termed "stable cavi-

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tion," or it may grow rapidly and collapse more or less violently, termed "inertial or transient cavitation." Neppiras¹⁰ and Apfel^{11,12} showed that the bubble behavior depends both on the bubble size compared to the acoustic wavelength and on the acoustic wave amplitude. Despite its fundamental interest, the estimation of these two parameters is a real problem in cavitation studies, even if acoustic amplitude may be determined with calibrated transducers. In fact, the cavitation state induced in the liquid is rarely described in most experimental works involving cavitation. So, easy monitoring methods have to be developed to correlate the cavitation state produced in a liquid to the chemical or biological effects observed. Some experimental methods for characterizing the cavitation bubble fields are described in the following paragraphs.

The easiest way to study cavitation may seem to be the direct observation of the bubble field. Because of the microscopic size and the rapid oscillations of each cavitation bubble, the direct observation of the phenomenon requires a very high spatial and temporal resolution photographic device. Lauterborn^{13–15} and his co-workers provided an important contribution to cavitation bubble studies by using high-speed photography. They also showed that holographic cinematography associated with digital processing of the three-dimensional images allows the cavitation bubble fields to be monitored. Unfortunately, the experimental device is too difficult to implement in industrial applications.

As cavitation bubble field visualization is quite complex, workers have used indirect observations of macroscopic criteria in order to describe the cavitation state produced in a liquid. The criterion selected to monitor for cavitation in industrial applications must combine real-time measurements on wide size ranges with easy installation and setup. We have considered most of the existing experimental techniques for cavitation detection. All of these techniques are based on the analysis of the interaction between the wave and the bubbly liquid. The optical solution, which consists of the analysis of the diffraction pattern obtained when a laser light propagates through a bubbly liquid, allows reliable, accurate, and nonintrusive bubble sizing.¹⁶ Unfortunately, we noticed that the sound wave which induces cavitation produces an acousto-optic effect. Thus, the diffraction pattern recorded results from both the interaction between the laser beam and the bubbles and the acousto-optic effect. It is therefore quite difficult to obtain information concerning an acoustic cavitation bubble field in this way. As it has been well-established that the presence of air bubbles in water can change the velocity and attenuation of an acoustic wave, we have considered an acoustical solution to characterize cavitation. Van Wijngaarden,¹⁷ and later Cafilish¹⁸ and Miksis,¹⁹ have modeled the transmission and scattering of an acoustic wave through a bubbly liquid. Moreover, it has been shown that the measurement of attenuation^{20–23} and velocity²⁴ of a low-power acoustic pulse propagating through a bubbly liquid provides quantitative information about the bubble field. However, this approach cannot be applied to the acoustic cavitation bubble field characterization because it assumes linear pulsations of stable bubbles about an equilibrium radius. Another acoustic solution to study cavitation has been

proposed by Roy^{25,26} and co-workers. The authors used a backscattering technique for the detection of transient cavitation. Signal processing applied to the backscattered wave could probably be used to monitor cavitation, but this method is very difficult to implement in industrial applications because of focusing and size constraints.

This list of methods has been limited to the easiest techniques. However, the discussion of their application to cavitation characterization for industrial applications shows that appropriate tools are needed to monitor cavitation. From this viewpoint, we have developed a cavitation probe based on the cavitation noise measurement. After a rapid review of different spectrum contributions present in the acoustic cavitation spectrum, we show that the spectral distribution of the cavitation noise is strongly linked to the cavitation state. Then, we propose the cavitation noise power, obtained by numerical integration of the spectra, as a new criterion to represent the cavitation state. We show that this criterion may be used to bring to the fore the hysteresis effect associated with cavitation phenomenon. Finally, we present the experimental correlation between this criterion and the sonochemical reactivity in an oxidization process.

II. BUBBLE OSCILLATIONS IN AN ACOUSTIC FIELD

In this study, we limit our interest to the case of a small bubble compared to the acoustic wavelength. Much work has been done concerning the dynamics of a single bubble subject to an acoustic wave. Noltingk and Neppiras²⁷ developed the first dynamic model of bubble oscillations. More complete models for the description of the dynamics of a spherical bubble have been proposed by Kirkwood and Bethe,²⁸ Gilmore,²⁹ and later by Prosperetti.³⁰ All of these models lead to a complex and nonlinear equation requiring a numerical solution. Prosperetti³¹ and Lauterborn³² have brought about a wide contribution to these computations. They have shown that the oscillating behavior of a bubble, which is strongly dependent on the acoustic wave amplitude and also on the relationship between the bubble size and the acoustic frequency, may be characterized by the spectrum of the radius–time function.

Whatever the model considered, the numerical computation of the radius–time variation shows that the frequency distribution of the oscillating motion may contain harmonics, subharmonics, and ultraharmonics of the driving frequency. The harmonic components arise from the strongly nonlinear motion of the bubble subject to forced oscillations. In the next two sections, we discuss the situations leading to subharmonic and/or ultraharmonic oscillations of the bubble. These occurrences differ essentially in terms of the acoustic wave intensity.

A. Subharmonic oscillations at low intensities

At low intensities, subharmonic oscillations result from bubbles that are twice the resonance size. These bubbles may present parametric oscillations for weak variations of their radius R around the initial value R_0 .

This phenomenon, already mentioned by Flynn³³ and Mei,³⁴ may be described by a simple analytical model. Con-

sider the Rayleigh–Plesset–Noltingk and Neppiras equation

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 + \frac{1}{\rho_0} \left[P_0 - P_A \sin \omega t + \frac{2\sigma}{R} - \left(P_0 + \frac{2\sigma}{R_0} \right) \left(\frac{R_0}{R} \right)^{3n} \right] = 0. \quad (1)$$

For small amplitude radial oscillations, we may write

$$R = R_0(1 + x), \quad (2)$$

where $x \ll 1$ and has the following form: $x = a \exp(j\omega t)$, where $a \ll 1$. So,

$$\dot{R} = R_0 \dot{x}; \quad \ddot{R} = R_0 \ddot{x}; \quad \frac{1}{R} \cong R_0(1 - x); \quad \left(\frac{R_0}{R} \right)^{3n} \cong 1 - 3nx.$$

Thus, motion equation (1) becomes

$$R_0^2 \ddot{x} + R_0^2 x \ddot{x} + \frac{3}{2} R_0^2 \dot{x}^2 + \frac{1}{\rho_0} \left[P_0 - P_A \sin \omega t + \frac{2\sigma}{R_0} (1 - x) - \left(P_0 + \frac{2\sigma}{R_0} \right) (1 - 3nx) \right] = 0. \quad (3)$$

Equation (3) may also be written

$$\ddot{x} + x \ddot{x} + \frac{3}{2} \dot{x}^2 + \frac{1}{\rho_0 R_0^2} \left[-P_A \sin \omega t + x \left\{ 3n \left(P_0 + \frac{2\sigma}{R_0} \right) - \frac{2\sigma}{R_0} \right\} \right] = 0. \quad (4)$$

Writing

$$\Omega_0^2 = \left[3n \left(P_0 + \frac{2\sigma}{R_0} \right) - \frac{2\sigma}{R_0} \right] \frac{1}{\rho_0 R_0^2}, \quad (5)$$

and

$$\Omega_0^2 h = \frac{P_A}{\rho_0 R_0^2}, \quad (6)$$

Eq. (4) is then written

$$\ddot{x} + x \ddot{x} + \frac{3}{2} \dot{x}^2 + \Omega_0^2 x = \Omega_0^2 h \sin \omega t. \quad (7)$$

Equation (5) reduces to Minnaert's expression³⁵ for small σ and $n = \gamma$.

Neglecting the $\dot{x}^2$ term and dividing by $(1 + x)$ factor with the approximation $1/(1 + x) \cong 1 - x$, Eq. (7) becomes

$$\ddot{x} + \Omega_0^2 x = \Omega_0^2 h (1 - x) \sin \omega t, \quad (8)$$

or

$$\ddot{x} + \Omega_0^2 (1 + h \sin \omega t) x = \Omega_0^2 h \sin \omega t. \quad (9)$$

The first member of Eq. (9) corresponds to the equation of a parametric oscillator described by Landau and Lifchitz.³⁶

This fact explains that in an acoustic cavitation field, bubbles having a R_0 radius corresponding to a resonance pulsation $\Omega_0 = \omega/2$, with ω the driving pulsation, will present a strong oscillating component at the $f/2$ subharmonic.

The parametric oscillation amplitude increases exponentially in function of time with an amplification coefficient $s = h\Omega_0/4$ given by Landau and Lifchitz³⁶ in the precise case of the half harmonic.

In spite of the drastic approximations made in the previous calculation, we may verify that an expression in the following form $ae^{+st} \cos(\Omega_0 t + \varphi)$ is a solution of expression (7) directly derived from the Rayleigh–Plesset–Noltingk–Neppiras equation (1) for low intensities.

Introduction of such a solution in relation (7) gives an s value equal to $s = h\Omega_0/12$. This amplification coefficient is lower than the previous one due to the damping effect ($\dot{x}^2$ term) taken into account in the latter case.

In conclusion, this result shows that the subharmonic oscillation appears even at low intensities in a randomly distributed cavitation bubble field, when bubbles are twice the resonance size for a driving frequency f . So, this component cannot be considered a significant parameter of a particular activity of cavitation bubbles.

B. Subharmonic and ultraharmonic oscillations at high intensities

Using methods of chaos physics to study the problem, Lauterborn³⁷ has described stochastic oscillations of a bubble. In this case, the spectral distribution of the radius–time function, obtained by Fourier transform, contains both subharmonic and ultraharmonic components. Each bubble of the cavitation -field may present chaotic oscillations for sufficiently high amplitude of the acoustic excitation. The higher the amplitude of the acoustic wave, the more ultraharmonic oscillations occur.

C. Cavitation noise induced by bubble oscillations

As chemical or biological effects induced by a bubble on the liquid are likely to be strongly linked to its oscillating behavior, it is paramount to evaluate this parameter. From an industrial viewpoint of cavitation applications, this problem is of fundamental interest. Spectral analysis of the cavitation noise should be a solution. In fact, each bubble of the cavitation field behaves as a secondary source of acoustic emission. So, the characteristics of each secondary source depends on the bubble motion. Llyichev^{38,39} has characterized the pressure wave emitted by a bubble as a function of the driving pressure. He has concluded that the spectrum of this pressure wave may contain harmonics, subharmonics, and ultraharmonics of the driving frequency as well as the oscillating motion of the bubble. These theoretical results confirm the experimental observations achieved by Neppiras.¹⁰

In the following parts, we will show that the cavitation noise spectrum analysis is efficient to characterize the cavitation activity. For example, the previous considerations have shown that the half-subharmonic emission may arise from two very different phenomena. So, this criterion, currently used as a cavitation probe, is ambiguous and a lot of work is still needed to distinguish the contribution of parametric oscillations of twice the resonant size bubbles from the contribution of chaotic oscillations to the subharmonic emissions. Because of the exponential increase in oscillation amplitude, the lifetime of twice the resonant size bubbles is very short in the liquid and their subharmonic emission at $f/2$ probably consists of short bursts. Conversely, the subharmonic and ultraharmonic emissions due to chaotic oscillations

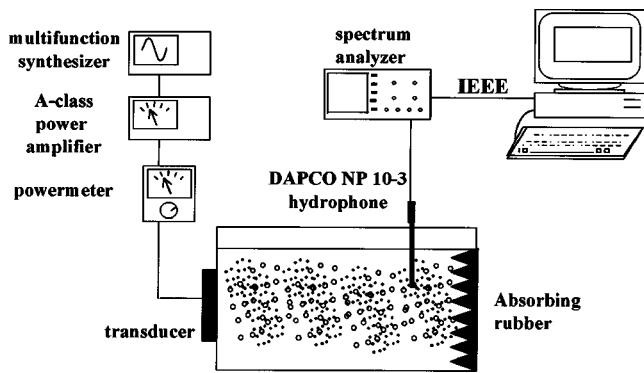


FIG. 1. Experimental setup.

tions of bubbles seem to be stable in time. This report should be a starting point for further investigations. In the meantime, in this paper we propose a new tool to characterize both the cavitation state and the cavitation activity in the liquid.

III. EXPERIMENTAL SETUP AND RESULTS

As a first step, we have observed the cavitation noise spectrum distortion as a function of the acoustic intensity. This value is deduced from the measured electrical power, knowing the electromechanical conversion factor of the transducer determined by a back-reflection technique.⁴⁰ The experimental setup is presented in Fig. 1. The liquid used is tap water saturated with air, at 20 °C and at atmospheric pressure. The transducer consists of a 6×6-cm piezoelectric transducer (PZT) ceramic bonded to a stainless-steel substrate. It is driven at 1.075 MHz. The cavitation noise is recorded with a broadband hydrophone DAPCO NP 10-3, and this signal is transmitted to an analogical spectrum analyzer, digitized, and stored in a computer. Cavitation noise spectrum is expressed in a logarithmic scale.

The results are presented in Fig. 2. These observations are quite similar to Esche.⁴¹ The cavitation noise spectrum at low acoustic intensities, typically under 24 mW/cm², is a line spectrum containing a fundamental line and harmonic, subharmonic, and ultraharmonic components. At high intensities, a broadband component is superimposed on this line spectrum. This signal, called “white noise” by Neppiras,¹⁰ may be attributed to two phenomena: strong chaotic oscillations of bubbles and shock waves emission by collapsing bubbles.

Llyichev^{38,39} has shown that ultraharmonic contributions from the bubble’s oscillating motion increase considerably at high intensities. Lauterborn³² has shown that all the spectral components broaden when intensity rises. Furthermore, Cordry⁴² and Crum have proposed another possibility, the emission of a shock wave by a single bubble. During the collapse, the bubble wall velocity is so high that a shock wave is released in the liquid. The shock wave produces a broadband contribution to the cavitation noise. The large amount of theoretical studies, as well as the observations of the motion of a single bubble in a standing wave field by Gaitan *et al.*⁴³ have greatly contributed to an understanding

of the oscillating behavior of each bubble. Then, it is possible to associate a particular motion of the bubble with each spectral component of the cavitation noise.

In Figs. 3 and 4, we show the $f/2$ and $2f$ spectrum component evolution measurements for different acoustic power. We can observe a sudden increase in these components for low intensities up to 60 mW/cm² in our experimental conditions. These results are in agreement with these of Neppiras.¹⁰ For higher acoustic power values, a threshold appears. These $f/2$ and $2f$ peaks show parametric or nonlinear bubble oscillations to be found in stable cavitation. They do not represent by themselves an indicator of transient cavitation state. In fact, $f/2$ and $2f$ peak levels reach a quasi-saturation value from 60 mW/cm² acoustic intensity, while we can see in Fig. 2 that noise contribution continues to increase significantly up to 120 mW/cm². The transient cavitation and bubble collapses produce white noise. The qualitative observation of the spectra shows the emergence and the increase in the cavitation noise with the acoustic intensity. So, it would be interesting to have a quantitative indicator of the transient cavitation correlated to the bubble collapses and to the cavitation bubble activity.

There are two possibilities to assess this overall contribution. The first way would consist of measuring a significant physical parameter by a cavitation spectrum integration in a linear scale. Then, we should obtain an estimation of the real cavitation noise power value. But, in this case, the anharmonic noise weakly contributes to the final result. The second way consists of a cavitation spectrum integration in a logarithmic scale in order to enhance the white-noise contribution. Although this integration does not really correspond to a physical power we will consider it, in the following, as a cavitation noise power indicator called CNP. The resulting data are linked to the cavitation noise power⁴⁴ within a frequency span corresponding to the hydrophone bandwidth. The result is presented in Fig. 5. This curve confirms the qualitative observations and the 120-mW/cm² characteristic value previously mentioned. We also observe that the cavitation noise power reaches a maximum value P_L . In Fig. 6 we present the respective noise and peak contributions to the CNP obtained by cutting the acoustic spectrum at a 30-dB level value corresponding to the maximum value of the white noise (see Fig. 2). The part over 30 dB has been considered as the peak contribution and the part below this value has been considered as the noise contribution. We may conclude from the comparison of both parts that the CNP is an appropriate indicator of the white noise since the subharmonic, harmonic, and ultraharmonic components contribute less than 10% to the CNP value.

In conclusion we may consider that the $f/2$ and $2f$ peak levels are good tools for the setting up of stable cavitation regime while the CNP is a good indicator to characterize transient cavitation state. The respective acoustic intensity thresholds are, in our case, around 60 and 120 mW/cm². Furthermore we have noticed a maximum value of CNP. In the next parts of this paper, we will normalize the cavitation noise power measurements in comparison with the limit value P_L . Using this new probe, we have studied two phenomena induced by acoustic cavitation in a liquid: a hys-

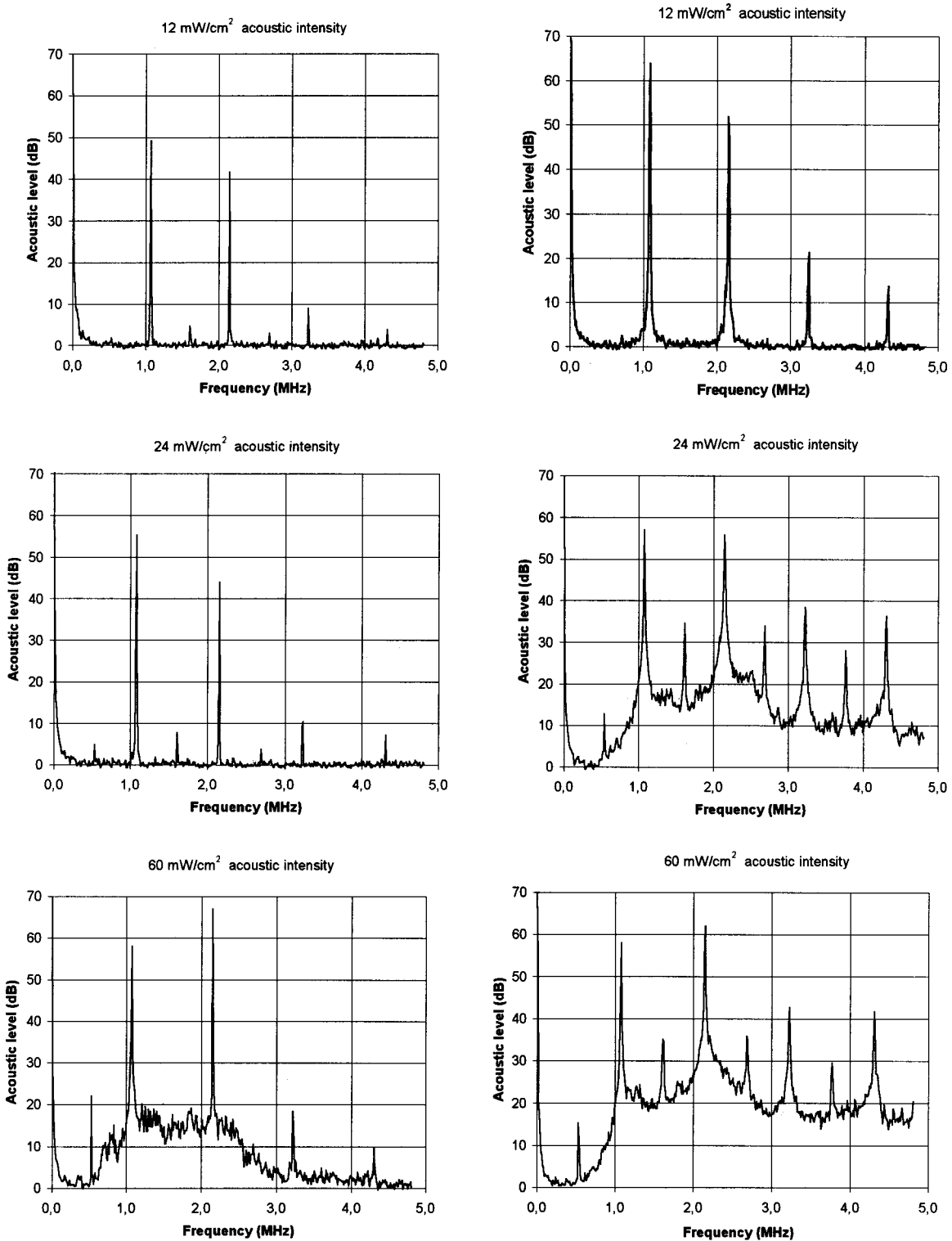


FIG. 2. Cavitation noise spectrum distortion as a function of the acoustic intensity transmitted to the liquid, for increasing power (left curves) and decreasing power (right curves).

esis effect linked to the increase in cavitation nuclei stabilized in the liquid and the sonolysis of water induced by free-radicals formation.

IV. HYSTERESIS EFFECT

Neppiras¹⁰ has shown that the cavitation state induced in a liquid depends on the sample history. Our observations of

$f/2$ and $2f$ spectrum component levels do not permit us to conclude that there is a significant hysteresis effect (Figs. 3 and 4). But, Fig. 2 clearly indicates a huge difference in acoustic spectra with regards to their history. We can also observe that spectral components broaden when acoustic power increases. So, we confirm the Lauterborn results. But, spectral component widening is maintained when acoustic power decreases. In order to detect the hysteresis phenom-

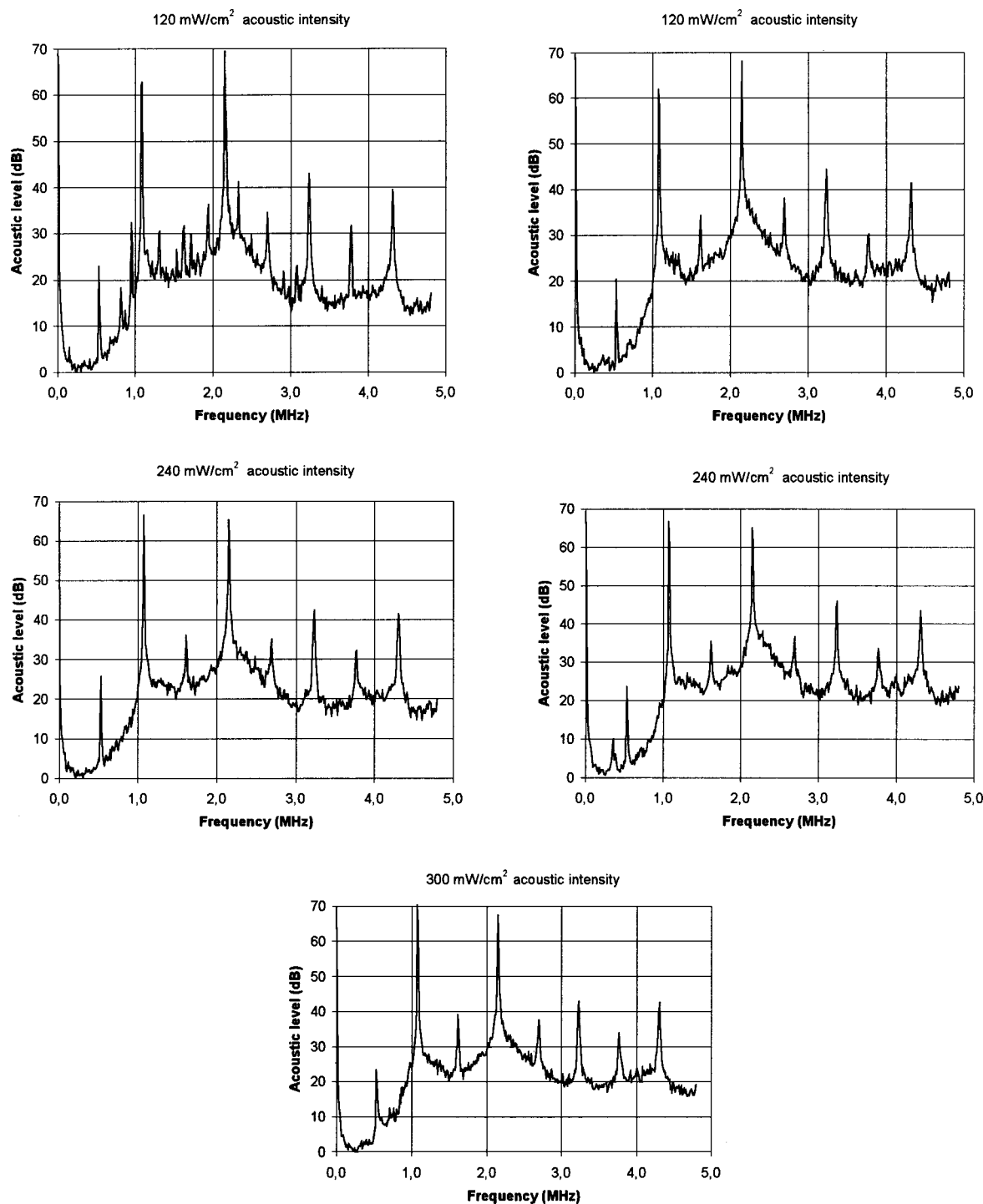


FIG. 2. (Continued.)

enon, we have recorded the cavitation noise power during an increasing power sweep, and immediately after a decreasing power sweep. The result obtained at a 1.075-MHz frequency is presented in Fig. 7. This curve clearly confirms the hysteresis effect, and shows that once generated in the liquid, a cavitation state needs a lower driving power to be maintained. This result might present a real industrial interest when this phenomenon takes place in the case of a large liquid volume locally insonified. We will check this point later.

V. SONOLYSIS STUDY

Cavitation bubble collapse induces free-radical generation in the liquid. Chemical effects of cavitation arise from this phenomenon. As the lifetime of these chemical species is very short, it is therefore very difficult to measure their concentration directly. So, their formation is brought to the fore using an oxidation process. We have established an experimental correlation between the cavitation noise power and the chemical reactivity in the $\text{Fe}^{2+}/\text{Fe}^{3+}$ oxidation reaction.⁴⁵ The driving frequency is 1.6 MHz. The cavitation

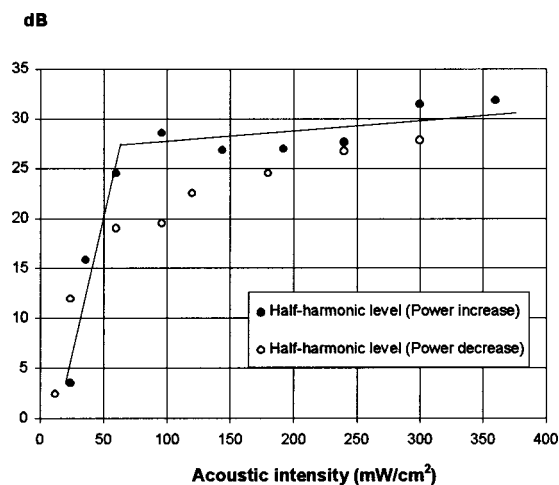


FIG. 3. Half-harmonic evolution as a function of the acoustic intensity transmitted to the liquid.

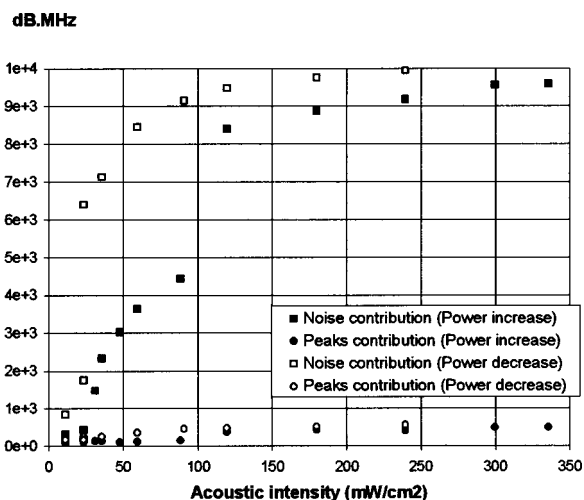


FIG. 6. Comparison between noise and peak contributions to the cavitation noise power.

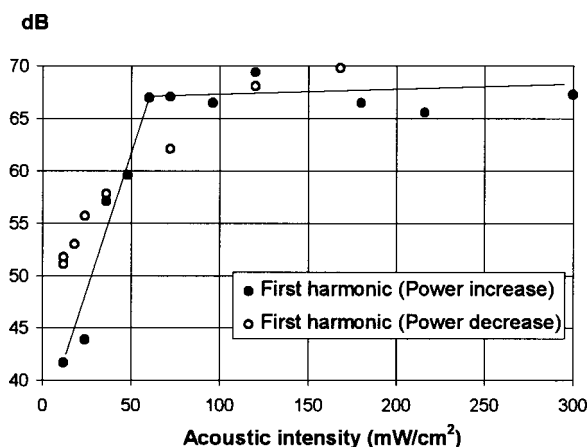


FIG. 4. First harmonic evolution as a function of the acoustic intensity transmitted to the liquid.

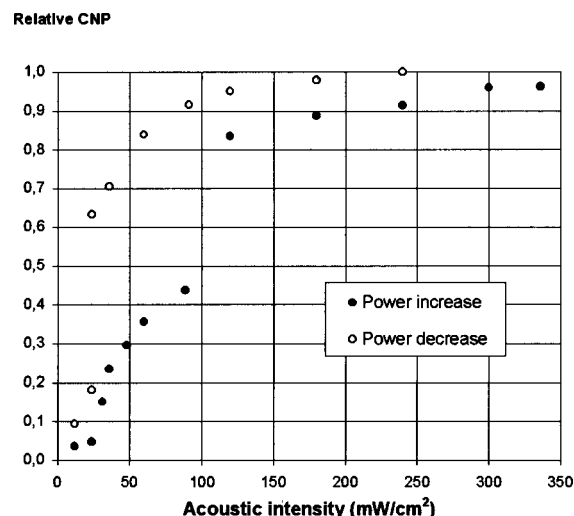


FIG. 7. Cavitation noise power as a function of the acoustic power transmitted to the liquid for increasing power sweep and for decreasing power sweep.

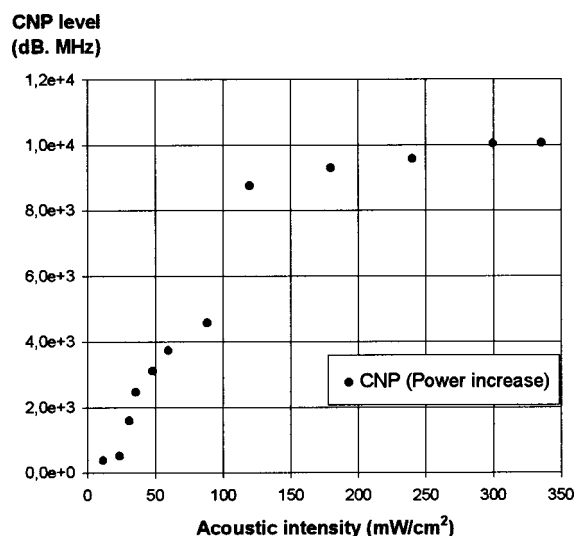


FIG. 5. Noise contribution to the cavitation noise power emission.

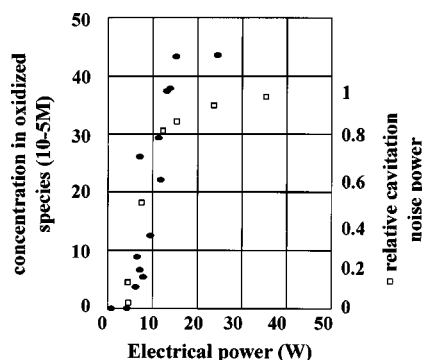


FIG. 8. Experimental correlation between acoustic cavitation noise power and sonochemical activity in an oxidation process.

noise is recorded with a DAPCO NP 10-3 hydrophone. The sonicated volume is 25 ml. The initial solution is $2.2 \times 10^{-2} \text{ M}(\text{Fe}^{2+})$. Figure 8 represents the concentration of oxidized species after a 30-min sonication for different driving intensities. We have noticed a very good agreement between the cavitation noise measured and the oxidized species concentration variation as a function of the driving power. For our specific sample, we have found that the triggering of sonochemical reactivity is closely linked to the emergence of white noise in the cavitation noise spectrum.

VI. CONCLUSION

In this paper, we have first presented a simple model which allows one to link the $f/2$ oscillations to the well-sized bubble parametric resonance. Then, we have shown that in order to correctly characterize the complex cavitation phenomenon, the stable cavitation regime should be distinguished from the transient cavitation state. Some parameters such as $f/2$ and $2f$ peak levels are well-suited to characterize the first situation. We present a new tool adapted to identify the second one. Furthermore, due to the fluctuations induced by the chaotic behavior of the bubbles, the peak levels are more difficult to appreciate than the overall contribution of the cavitation noise. This powerful tool allows both cavitation characterization and sonochemical reactivity overview. Moreover, this new criterion combines real-time measurements and easy implementation. We have validated this probe in the case of an oxidation process and we have used it to show the hysteresis effect of cavitation. So, this method is suitable for cavitation characterization within industrial applications. However, further work will consist of applying this new tool to different industrial cavitation applications and to check if the previous results obtained near 1 MHz are valid in another frequency ranges.

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